Garrett Comes

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Education

Georgia Institute of Technology **Jul 2022 – Aug 2026**
B.S. Aerospace Engineering, Concentration in Propulsion *Atlanta, GA*

- Coursework: Jet & Rocket Propulsion, Computational Fluid Dynamics, Aerothermodynamics, Finite Element Analysis, Control System Design, Aircraft Flight Dynamics, Aerodynamics.

Experience

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space **Summers 2022 – 2025**
Flight programs incl. Dragonfly and Mars Sample Return *Waterton Campus, Denver, CO*

- Designed a piston-tank propellant gauging system from a blank sheet: internal hermetically sealed magnets sensed by radiation-hardened Hall-effect sensors; owned magnetic circuit sizing, sensor selection, and integration concept.
- Built propellant slosh CFD simulations of spacecraft tanks in Ansys and supported design of a slosh characterization test rig used to correlate model predictions against test data.
- Contributed to a green-propellant trade study evaluating hydrazine-replacement monopropellants across performance, materials compatibility, and handling/safety criteria.
- Designed and analyzed propulsion hardware and fluid-system components, incl. valves and test support equipment, with Creo CAD models and drawings; supported test campaigns, requirements verification, and anomaly investigation.

Projects & Research

SCRAP — Orbital Debris Detumbling & Capture, Senior Design | *Python, FEniCSx, Rust, Gmsh* **2025 – 2026**

- Co-authoring an AIAA SciTech 2027 extended abstract on a servicing spacecraft that detumbles and captures orbital debris with a robot-arm-mounted YBCO high-temperature-superconducting coil.
- Developed a coupled electromagnetic–thermal quench simulation of the HTS pancake coil, using an H-formulation with E–J power-law resistivity, per-turn tape meshing, and radiation/cold-finger BCs, to establish safe operating current margins.
- Built spacedb, a Rust space-object database and GUI: collision and decay analysis, target value scoring, and multi-target servicing route optimization over the full public catalog.
- Trained DDPG reinforcement-learning agents for eddy-current detumbling of uncooperative targets.

Valurile — GPU Lattice-Boltzmann CFD Solver | *Rust, CUDA* **2024 – present**

- Wrote a multi-crate LBM solver implementing Esoteric Pull in-place streaming, roughly halving memory traffic per timestep, plus static mesh refinement, Smagorinsky–Lilly LES, and SRT/TRT/MRT collision operators.
- Built an interactive egui/GPU frontend: airfoil and cylinder obstacles, physical-to-lattice nondimensionalization, live velocity/density/pressure/vorticity fields, and drag/lift coefficient histories.

Thermodynamic Sampling Unit Research | *Python, JAX, PyTorch* **2026**

- Hybridized a pretrained masked-diffusion language model with block-Gibbs sampling on Extropic's THRML hardware simulator; joint factor-graph decoding beat independent argmax by 7–15 pp on held-out data.
- Quantified sampling coverage/energy frontiers across 59 randomized Potts landscapes with a bit-for-bit reproducible pipeline, 9/9 regression tests passing, and adversarial review gates on all headline claims.

fem_rust_ae — Finite-Element Solver in Rust, live in the browser | *Rust, WASM* **2025**

- Implemented an FEM structural solver from scratch with mesh convergence studies; compiled to WebAssembly for an interactive in-browser demo.

Skills

Languages: Python, Rust, C++, MATLAB, CUDA/Triton, LaTeX
Simulation & CAE: Ansys Fluent and Mechanical, FEniCSx, Gmsh, SolidWorks, Creo, Fusion 360, Inventor
Scientific computing: JAX, PyTorch, NumPy/SciPy, wgpu, MPI/PETSc, engineering optimization
Tooling: Linux/NixOS, Git, agentic AI development with Claude Code, MCP servers, and custom agent frameworks

Leadership & Certifications

Yellow Jacket Space Program, 2023 – present: pressurant isolation valve design for a collegiate liquid-propellant space-shot vehicle.
Certifications: SolidWorks CSWA · Autodesk Inventor · Autodesk Fusion 360 · Wolfram Language Level 1